

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 2 Summer 2025 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 2

Subject Code & Name: 24AF1000ES208 - BASIC CIVIL AND MECHANICAL ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 What is the primary purpose of a foundation in a building?
 - a. To provide aesthetic appeal
 - b. To distribute the loads to the ground
 - c. To create storage space
 - d. To improve ventilation
- 2 What is the primary function of aggregate in concrete ?
 - a. To provide strength
 - b. To improve workability
 - c. To reduce shrinkage
 - d. To increase volume
- 3 ----- consists of a group of member which form a support for a doors.
 - a. Sill
 - b. Style
 - c. Frame
 - d. Head
- 4 A ----- is a horizontal member which are placed across an opening to support the portion of the structure above it.
 - a. Lintel
 - b. Windows
 - c. Sill
 - d. Doors
- 5 ----- is used to find the elevations of given points with respect to the ground.
 - a. Plane Table surveying
 - b. Traversing
 - c. Levelling
 - d. Contouring

- 6 Benchmark is defined as ----- a.
a. fixed points of known RL
b. Nonfixed point
c. RL is not known
d. None
- 7 Vertical distance between any two consecutive contours is called as
a. Vertical equivalent
b. Contour interval
c. Horizontal equivalent
d. All
- 8 Which of the following kind of energy output is obtained from a nuclear power plant
a. Electricity
b. Thermal Energy
c. nuclear energy
d. Heat energy
- 9 Diesel Engine is ----- than petrol engine of same capacity.
a. Lighter
b. Heavier
c. Unpredictable
d. None
- 10 Pattern is required in -----
a. Sand mould casting
b. Permanent mould casting
c. Die casting
d. None
- 11 Which of the following operation is not carried out on drilling machine
a. Boring
b. Tapping
c. Reaming
d. Facing
- 12 Engineering applications of steady flow energy equations are in
a. Engines & Turbines
b. Pumps & Compressor
c. Nossels & Heat Exchanger
d. All
- Q2. Solve the following.
- A) Describe in detail mortar along with its types & properties. 6
- B) Explain the brick as a construction material along with its properties & uses. 6
- Q3. Solve the following.
- A) Define Foundation with its types & functions 6
- B) What are the requirement of good stair ? 6
- Q4. Solve Any Two of the following.
- A) Explain Chain surveying & mention its types & Describe in detail metric chain with sketch. 6
- B) What are the advantages & disadvantages of Plane table surveying 6

- C) Define the terms i) Back sight ii) Fore sight iii) Reduced Level iv) Bench mark v) Contour map vi) Contour line 6

Q5. Solve Any Two of the following.

- A) Describe in brief second law of Thermodynamics with the help of a) Kelvin-Planck statement b) Clausius Statement 6
- B) Explain the working of four stroke petrol engine with suitable diagram. 6
- C) With suitable diagram, explain the construction & working of Nuclear power plant. 6

Q6. Solve Any Two of the following.

- A) Describe any two mechanisms with suitable sketch a) Crank rocker b) Double crank c) Reciprocating engine 6
- B) What are the factors affecting mechanical properties of Engineering materials? Enlist the various types of cast iron along with their applications. 6
- C) With neat sketch, explain the construction of Lathe machine. 6

*** End ***